Эволюция

Основа: загруженный учебник Biology Grade 9, PDF-страницы 316–347. Оригинальные страницы с иллюстрациями приложены в конце.

Как заниматься

Прочитай объяснения, изучи соответствующие схемы на оригинальных страницах, ответь на пять вопросов после каждой темы. Затем выполни итоговую проверку из 40 вопросов. Проходной ориентир — 28 правильных ответов из 40.

16.1 The History of Life on Earth

16.1 The History of Life on Earth You will: study all the basic stages of development of life on Earth. Key terms Эон - период в истории Земли; Эра - подинтервал эона; Ископаемые - останки организмов. STQ What type of organisms lived on our planet millions or billions years ago? Text According to the fossils, the oldest living organisms had lived 3.5 billion of years ago. They were primitive prokaryotic unicellular microorganisms. Over last 3.5 billion years, life changed and developed, producing a large diversity of complex and highly adapted organisms. Scientists study fossils to understand when and how living organisms changed. Using radiometric dating technique, they can define exact time of any fossil. The time of Earth is divided into periods called eons. Some eons are divided into eras. The history of life on Earth: 1. Archean eon (4 to 2.5 billion years)

1. the beginning of photosynthesis; 2. living organisms begin to move on land; 3. the appearance of the first multicellular organisms; 4. the first mammals; 5. the first humans. Facts The origin of Life There are no proved theories about Origin of Life on our planet. But there are some ideas: “Panspermia” theory says that life on Earth came from space; “Creationism” says that all living things were created by God; “Biochemical Evolution” says that living things were spontaneously formed from different molecules. Research time Ученые считают, что птицы являются потомками динозавров. Можете ли вы объяснить, почему? Literacy 1. Scientists discovered that the amount of O2 in the atmosphere rapidly increased in the period from 2.7 to 2.4 billions years ago. Explain the reasons for this event? 2. Scientists say that the spread of angiosperm plants and insects in Cenozoic era are related. Explain why? Terminology adapted - бейімделген / адаптированный; algae - балдыр / водоросли;

Пять вопросов для самопроверки

1. Объясни своими словами, что означает «The History of Life on Earth».

2. Назови три ключевых понятия или структуры, связанные с темой «The History of Life on Earth».

3. Какой пример из учебника иллюстрирует эту тему?

4. Какая схема, таблица или иллюстрация помогает понять раздел 16.1? Объясни её.

5. Сформулируй один вывод по теме 16.1 и обоснуй его.

16.2 The Theories of Evolution

16.2 The Theories of Evolution You will: study the ideas of Linnaeus and Lamarck; explain the role of Charles Darwin in the creation of the Theory of Evolution. Key terms Эволюция - изменение групп организмов в течение длительного периода; Теория - объяснение различных фактов и наблюдений. STQ Where did all living creatures come from? Text There are many living organisms. People always tried to explain how all living things appeared. Some scientists thought that all species are constant, they are not changing, and there are no new species. One of them was Carolus Linnaeus, a Swedish botanist, who believed that all organisms were created by God. He made “Classification system”. In this system,

organisms are grouped according to their characteristics. Other scientists thought that organisms slowly change over long periods of time, creating new species. This is called evolution. One of them was French biologist Jean-Baptiste Lamarck (1744 – 1829). He thought that if an organism uses one of its organs more often than others, this organ in the next

Research time Если бы теория Ламарка была правильной, как изменилась бы ваша жизнь? Literacy 1. Explain why the theory of Lamarck is wrong. 2. What type of scientific evidence do evolutionists use to support Darwinism? 3. The Classification system was created by Linnaeus, but then Darwin used it to prove his theory. How did Darwin use Classification system? Terminology according to – сәйкес / в соответствии с; adapted – бейімделген / адаптированный; environment – қоршаған орта / окружающая среда; finch - шұбар шымшық / вьюрок; make notes – жазып алу / делать записи; observation – байқау / наблюдение; related – байланысқан / связанный; species - түр / вид; to appear - пайда болу / появляться; to notice – байқау / замечать; turn out – болып қалу / оказаться; voyage – саяхат / путешествие.

Пять вопросов для самопроверки

1. Объясни своими словами, что означает «The Theories of Evolution».

2. Назови три ключевых понятия или структуры, связанные с темой «The Theories of Evolution».

3. Какой пример из учебника иллюстрирует эту тему?

4. Какая схема, таблица или иллюстрация помогает понять раздел 16.2? Объясни её.

5. Сформулируй один вывод по теме 16.2 и обоснуй его.

16.3 Darwinism

16.3 Darwinism You will: characterise the driving forces of evolution; describe the role of natural selection in adaptations. Key terms Естественный отбор - основной фактор эволюции, приводящий к выживанию и преимущественному размножению более приспособленных к данным условиям среды особей, обладающих полезными наследственными признаками; Адаптация - процесс приспособления организма к окружающей среде; Популяция – организмы одного вида, обитающие в виде групп; Разнообразие - различия между организмами в одной популяции. STQ Is evolution real? Text There are important points in Darwinism: 1. There are differences among organisms in one population: because of sexual reproduction, mutations, and environmental factors organisms are different in one population. This is called variation. 2. Organisms produce many offspring. Many of these new organisms will not survive. 3. Organisms with “better” (stronger or fittest) characteristics survive and produce offspring, while organisms with bad or weak characteristics

advantages to the cell. So, these incomplete mechanisms will not stay, they must be deleted. Variation is an important condition for evolution. Rotifers are animals which have reproduced only asexually for nearly 200 million years. They were all identical, and there was no variation among them. But they were evolving all this time! Scientists still search for explanation of Rotifers’ evolution. Research time Помимо естественного отбора также существует искусственный отбор. Попытайтесь объяснить, что это такое, и приведите примеры. Literacy 1. Inheritance is important for evolution. Explain why. 2. Explain disruptive selection. Terminology abundant - жайылған / распространённый; advantage - артықшылық / преимущество; among - арасында / среди; camouflage - бүркену / маскировка; competition - бәсекелестік / конкуренция; fittest - бейімделген / приспособленный; flounder - камбала; invisible - көзге түспейтін / невидимый; mimicry - мимикрия; offspring - ұрпақ / потомство; rotifera - коловраткалар / коловратки; struggle - күрес / борьба; to differ - айырмашылығы болу / отличаться; to mimic

Пять вопросов для самопроверки

1. Объясни своими словами, что означает «Darwinism».

2. Назови три ключевых понятия или структуры, связанные с темой «Darwinism».

3. Какой пример из учебника иллюстрирует эту тему?

4. Какая схема, таблица или иллюстрация помогает понять раздел 16.3? Объясни её.

5. Сформулируй один вывод по теме 16.3 и обоснуй его.

16.4 Speciation

16.4 Speciation You will: characterize the criteria of the species; explain the causes of speciation. Key terms Вид - группа схожих организмов, способная к взаимному скрещиванию; Видообразование - процесс возникновения новых биологических видов. STQ How to produce a new species? Text There are millions of different species on the Earth. But what is a species? Why a leopard and a jaguar are different species, while St. Bernard dog and Pekingese are the same? There are some criteria we use for organisms of one species: 1. Organisms of one species must have the same chromosome number in their cells. 2. Organisms of one species must be able to mate. 3. The offspring of two organisms of one species must be fertile. New species are formed in a process called speciation. During this process two groups of organisms of one population cannot reproduce with each

1. Find out allopatric speciation. Explain your answer. 2. Find out sympatric speciation. Explain your answer. 3. There is one more type of speciation called parapatric speciation. Using picture above explain parapatric speciation. Facts Intermediate species are evolutionary connection between two groups of organisms. For example, between fish and amphibia. According to evolution, they should have characteristics between previous group and emerging group. So, they would fail the competition with more adapted species. Mostly fossils of intermediate species are not found. Research time Найдите различные примеры видообразования. Определите тип видообразования и объясните почему.

Пять вопросов для самопроверки

1. Объясни своими словами, что означает «Speciation».

2. Назови три ключевых понятия или структуры, связанные с темой «Speciation».

3. Какой пример из учебника иллюстрирует эту тему?

4. Какая схема, таблица или иллюстрация помогает понять раздел 16.4? Объясни её.

5. Сформулируй один вывод по теме 16.4 и обоснуй его.

ИТОГОВАЯ ПРОВЕРКА · 40 ВОПРОСОВ

Ответь письменно. Проверяй ответы по соответствующим разделам и рисункам оригинального учебника. Это открытые задания; готового ключа с вариантами A–D здесь нет.

1. Дай определение ключевому понятию раздела «The History of Life on Earth».
2. Дай определение ключевому понятию раздела «The Theories of Evolution».
3. Дай определение ключевому понятию раздела «Darwinism».
4. Дай определение ключевому понятию раздела «Speciation».
5. Опиши основной процесс, рассматриваемый в разделе «The History of Life on Earth».
6. Опиши основной процесс, рассматриваемый в разделе «The Theories of Evolution».
7. Опиши основной процесс, рассматриваемый в разделе «Darwinism».
8. Опиши основной процесс, рассматриваемый в разделе «Speciation».
9. Приведи биологический пример по теме «The History of Life on Earth».
10. Приведи биологический пример по теме «The Theories of Evolution».
11. Приведи биологический пример по теме «Darwinism».
12. Приведи биологический пример по теме «Speciation».
13. Назови две причины или условия, влияющие на явления в теме «The History of Life on Earth».
14. Назови две причины или условия, влияющие на явления в теме «The Theories of Evolution».
15. Назови две причины или условия, влияющие на явления в теме «Darwinism».
16. Назови две причины или условия, влияющие на явления в теме «Speciation».
17. Сравни два объекта или процесса, встречающихся в разделе «The History of Life on Earth».
18. Сравни два объекта или процесса, встречающихся в разделе «The Theories of Evolution».
19. Сравни два объекта или процесса, встречающихся в разделе «Darwinism».
20. Сравни два объекта или процесса, встречающихся в разделе «Speciation».
21. Объясни, как связаны строение и функция в теме «The History of Life on Earth».
22. Объясни, как связаны строение и функция в теме «The Theories of Evolution».
23. Объясни, как связаны строение и функция в теме «Darwinism».
24. Объясни, как связаны строение и функция в теме «Speciation».
25. Какой вывод можно сделать из рисунков раздела «The History of Life on Earth»?
26. Какой вывод можно сделать из рисунков раздела «The Theories of Evolution»?
27. Какой вывод можно сделать из рисунков раздела «Darwinism»?
28. Какой вывод можно сделать из рисунков раздела «Speciation»?
29. Составь краткую схему по разделу «The History of Life on Earth».
30. Составь краткую схему по разделу «The Theories of Evolution».
31. Составь краткую схему по разделу «Darwinism».
32. Составь краткую схему по разделу «Speciation».
33. Какая ошибка чаще всего возникает при объяснении темы «The History of Life on Earth»? Исправь её.

34. Какая ошибка чаще всего возникает при объяснении темы «The Theories of Evolution»? Исправь её.

35. Какая ошибка чаще всего возникает при объяснении темы «Darwinism»? Исправь её.

36. Какая ошибка чаще всего возникает при объяснении темы «Speciation»? Исправь её.

37. Придумай практический вопрос по теме «The History of Life on Earth» и ответь на него.

38. Придумай практический вопрос по теме «The Theories of Evolution» и ответь на него.

39. Придумай практический вопрос по теме «Darwinism» и ответь на него.

40. Придумай практический вопрос по теме «Speciation» и ответь на него.

Критерий прохождения: 28/40 (70%). При открытых ответах оценивай каждый вопрос в 1 балл за корректный, содержательный ответ.

Оригинальные страницы учебника

Далее приведены страницы исходного PDF без изменений. На них находятся оригинальные иллюстрации, упражнения и таблицы.

Все учебники Казахстана на OKULYK.KZ

CHAPTER 16.0

Evolution



*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

16.1 The History of Life on Earth

You will:

- study all the basic stages of development of life on Earth.

Key terms

- Эон - период в истории Земли;
- Эра - подинтервал эона;
- Ископаемые - останки организмов.

STQ

What type of organisms lived on our planet millions or billions years ago?

Text

According to the fossils, the oldest living organisms had lived 3.5 billion of years ago. They were primitive prokaryotic unicellular microorganisms. Over last 3.5 billion years, life changed and developed, producing a large diversity of complex and highly adapted organisms.

Scientists study fossils to understand when and how living organisms changed. Using radiometric dating technique, they can define exact time of any fossil. The time of Earth is divided into periods called eons. Some eons are divided into eras.

The history of life on Earth:

1. Archean eon (4 to 2.5 billion years)

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

The unicellular prokaryotes first appeared 3,500 mya (million years ago). They were the only living things on Earth for next 1.5 billion years. Also in this eon, some organisms started to make photosynthesis.

2. Proterozoic eon (2,500 to 542 mya)

The first eukaryotes were found 1.8 billion years ago. Then, 1.2 billion years ago first multicellular organisms emerged, they were small red algae. At the end of this eon first primitive soft-bodied animals come into existence.
The last eon is a Phanerozoic eon. It is divided into three eras:

1. Paleozoic era (542-251 mya)

Colonization of land begins. Also, the origin of many groups of organisms happened, including many types of algae, vascular and seed plants, fish, insects, amphibians, reptiles and many others.

2. Mesozoic era (251-65 mya)

The origin of mammals at the beginning. Gymnosperm trees and dinosaurs are abundant most time of this era. In the end, dinosaurs became extinct, but angiosperms appeared.

3. Cenozoic era (65 mya - now)

Mammals, birds, angiosperms and insect spread on land. First primates come into existence. Ice age began 2.6 mya with mammoths and first humans.

Activity

Imagine you have a clock with time scale not for 12 hours, but for 3.5 billion years: from the first organism to this day. On this clock label each eon or era. Then mark these events:

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

1. the beginning of photosynthesis;
2. living organisms begin to move on land;
3. the appearance of the first multicellular organisms;
4. the first mammals;
5. the first humans.

Facts

The origin of Life

There are no proved theories about Origin of Life on our planet. But there are some ideas:

- “Panspermia” theory says that life on Earth came from space;
- “Creationism” says that all living things were created by God;
- “Biochemical Evolution” says that living things were spontaneously formed from different molecules.

Research time

Ученые считают, что птицы являются потомками динозавров. Можете ли вы объяснить, почему?

Literacy

1. Scientists discovered that the amount of O2 in the atmosphere rapidly increased in the period from 2.7 to 2.4 billions years ago. Explain the reasons for this event?
2. Scientists say that the spread of angiosperm plants and insects in Cenozoic era are related. Explain why?

Terminology

- adapted - бейімделген / адаптированный;
- algae - балдыр / водоросли;

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

- angiosperms - жабық тұқымдылар немесе гүлді өсімдіктер / покрытосеменные или цветковые;
- existence - тіршілік / существование;
- eon - эон;
- mammoth - мамонт;
- origin - туындау / возникновение;
- spontaneous - кенеттен / спонтанный;
- to appear/ to emerge - пайда болу / появляться;
- to become extinct - құрып біту / вымирать.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

16.2 The Theories of Evolution

You will:

- study the ideas of Linnaeus and Lamarck;
- explain the role of Charles Darwin in the creation of the Theory of Evolution.

Key terms

- Эволюция - изменение групп организмов в течение длительного периода;
- Теория - объяснение различных фактов и наблюдений.

STQ

Where did all living creatures come from?

Text

There are many living organisms. People always tried to explain how all living things appeared.

Some scientists thought that all species are constant, they are not changing, and there are no new species. One of them was Carolus Linnaeus, a Swedish botanist, who believed that all organisms were created by God. He made “Classification system”. In this system, organisms are grouped according to their characteristics.

Other scientists thought that organisms slowly change over long periods of time, creating new species. This is called evolution. One of them was French biologist Jean-Baptiste Lamarck (1744 – 1829). He thought that if an organism uses one of its organs more often than others, this organ in the next

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

generation will be slightly bigger or stronger. One example is a giraffe's neck. When a giraffe tries to eat leaves from higher branches of a tree, it stretches its neck. That's why giraffe's offspring will have a slightly longer neck.

Lamarck's theory (Lamarckism) turns out to be wrong. Another evolutionary theory was proposed by Charles Darwin, an English naturalist. At 1831, he started a long voyage around the world on a ship Beagle. During this voyage, he made notes about all animals and plants on different islands. For example, on Galapagos Islands, he studied different species of finches. These birds were similar, but there was a little difference among them. Darwin noticed that these differences were related to the differences in their food type. Other examples also show that organisms are highly adapted to their environment.
After all these observations, Darwin made his theory of Evolution (Darwinism), according to which species change over time, producing more adapted species.

Activity

Phylogenetic tree.

According to Darwinism, all organisms are related to each other: different groups of organisms originated at a different time from other groups. Look at this tree.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ



This is called Phylogenetic tree. It shows how different groups are related, and when they appeared. Write in the blank these animals to make a correct phylogenetic tree:

- cow;
- human;
- frog;
- shark;
- lancelet;
- ant;
- sparrow.

Facts

The origin of Life

There are no proved theories about Origin of Life on our planet. But there are some ideas:

- “Panspermia” theory says that life on Earth came from space;
- “Creationism” says that all living things were created by God;
- “Biochemical Evolution” says that living things were spontaneously formed from different molecules.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

Research time

Если бы теория Ламарка была правильной, как изменилась бы ваша жизнь?

Literacy

1. Explain why the theory of Lamarck is wrong.
2. What type of scientific evidence do evolutionists use to support Darwinism?
3. The Classification system was created by Linnaeus, but then Darwin used it to prove his theory. How did Darwin use Classification system?

Terminology

- according to – сәйкес / в соответствии с;
- adapted – бейімделген / адаптированный;
- environment – қоршаған орта / окружающая среда;
- finch - шұбар шымшық / вьюрок;
- make notes – жазып алу / делать записи;
- observation – байқау / наблюдение;
- related – байланысқан / связанный;
- species - түр / вид;
- to appear - пайда болу / появляться;
- to notice – байқау / замечать;
- turn out – болып қалу / оказаться;
- voyage – саяхат / путешествие.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

16.3 Darwinism

You will:

- characterise the driving forces of evolution;
- describe the role of natural selection in adaptations.

Key terms

- Естественный отбор - основной фактор эволюции, приводящий к выживанию и преимущественному размножению более приспособленных к данным условиям среды особей, обладающих полезными наследственными признаками;
- Адаптация - процесс приспособления организма к окружающей среде;
- Популяция – организмы одного вида, обитающие в виде групп;
- Разнообразие - различия между организмами в одной популяции.

STQ

Is evolution real?

Text

There are important points in Darwinism:

1. There are differences among organisms in one population: because of sexual reproduction, mutations, and environmental factors organisms are different in one population. This is called variation.
2. Organisms produce many offspring. Many of these new organisms will not survive.
3. Organisms with “better” (stronger or fittest) characteristics survive and produce offspring, while organisms with bad or weak characteristics

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

often die and do not reproduce. This is called Natural Selection.

4. And because of reproducing of “better” organisms, over many generations in the population will stay only the fittest organisms. So, it is a population that changes, not an organism.

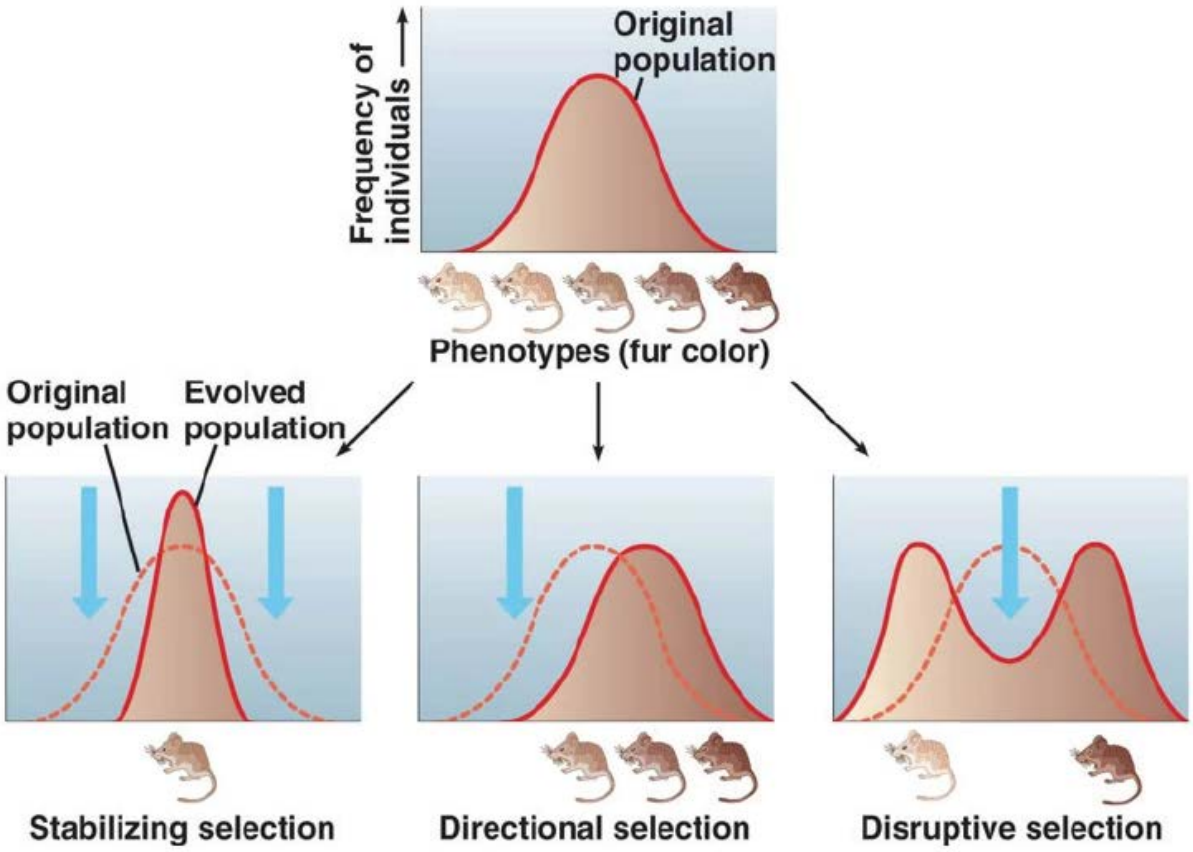


Natural selection: gray mice have an advantage over white mice. Over time there is no white mouse in this population.

There are different types of natural selection: directional, stabilising, and disruptive. In stabilising selection organisms with the most abundant characteristics has advantage over others and survive. Directional selection happens when environment changes and a few organisms have an advantage.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ



Three types of natural selection

As natural selection works the fittest organisms stay and population changes. Over time these changes produce adapted organisms. Examples of adaptations are camouflage and mimicry.

Camouflage is when an organism has colours that make it invisible in its environment. Examples are zebras, chameleon, flounder and many others.

Mimicry is when an organism of one species looks like (mimics) an organism of another species. Animals often mimics stronger or more dangerous (maybe poisonous) species for protection. Most of the dangerous species have bright warning colours. These colours are signals to others which means “don’t touch me!”.

Activity

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

You can see moth of different colours on this picture.



These insects are mostly found on trees. Imagine these conditions and answer to the questions:

1. Trees are white. Explain how and why population will change.
2. Then, trees become black. Explain how and why population changes now.

Facts

- Different molecular mechanisms in the cell consist of many components. Without any of this component, mechanisms will not work. According to evolution, all components can not emerge altogether at the same time. Instead, they should appear one by one in a long period. In this case, they will not work, and will not give any

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

advantages to the cell. So, these incomplete mechanisms will not stay, they must be deleted.
- Variation is an important condition for evolution. Rotifers are animals which have reproduced only asexually for nearly 200 million years. They were all identical, and there was no variation among them. But they were evolving all this time! Scientists still search for explanation of Rotifers' evolution.

Research time

Помимо естественного отбора также существует искусственный отбор. Попытайтесь объяснить, что это такое, и приведите примеры.

Literacy

1. Inheritance is important for evolution. Explain why.
2. Explain disruptive selection.

Terminology

- abundant - жайылған / распространённый;
- advantage - артықшылық / преимущество;
- among - арасында / среди;
- camouflage - бүркену / маскировка;
- competition - бәсекелестік / конкуренция;
- fittest - бейімделген / приспособленный;
- flounder - камбала;
- invisible - көзге түспейтін / невидимый;
- mimicry - мимикрия;
- offspring - ұрпақ / потомство;
- rotifera - коловраткалар / коловратки;
- struggle - күрес / борьба;
- to differ - айырмашылығы болу / отличаться;
- to mimic - еліктеу / подражать;
- to survive - тірі қалу / выживать;
- warning - ескерту / предупреждение.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

16.4 Speciation

You will:

- characterize the criteria of the species;
- explain the causes of speciation.

Key terms

- Вид - группа схожих организмов, способная к взаимному скрещиванию;
- Видообразование - процесс возникновения новых биологических видов.

STQ

How to produce a new species?

Text

There are millions of different species on the Earth. But what is a species? Why a leopard and a jaguar are different species, while St. Bernard dog and Pekingese are the same?

There are some criteria we use for organisms of one species:

1. Organisms of one species must have the same chromosome number in their cells.
2. Organisms of one species must be able to mate.
3. The offspring of two organisms of one species must be fertile.

New species are formed in a process called speciation. During this process two groups of organisms of one population cannot reproduce with each

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

other. This is called reproductive isolation. After a long period of time these organisms change and form new species.

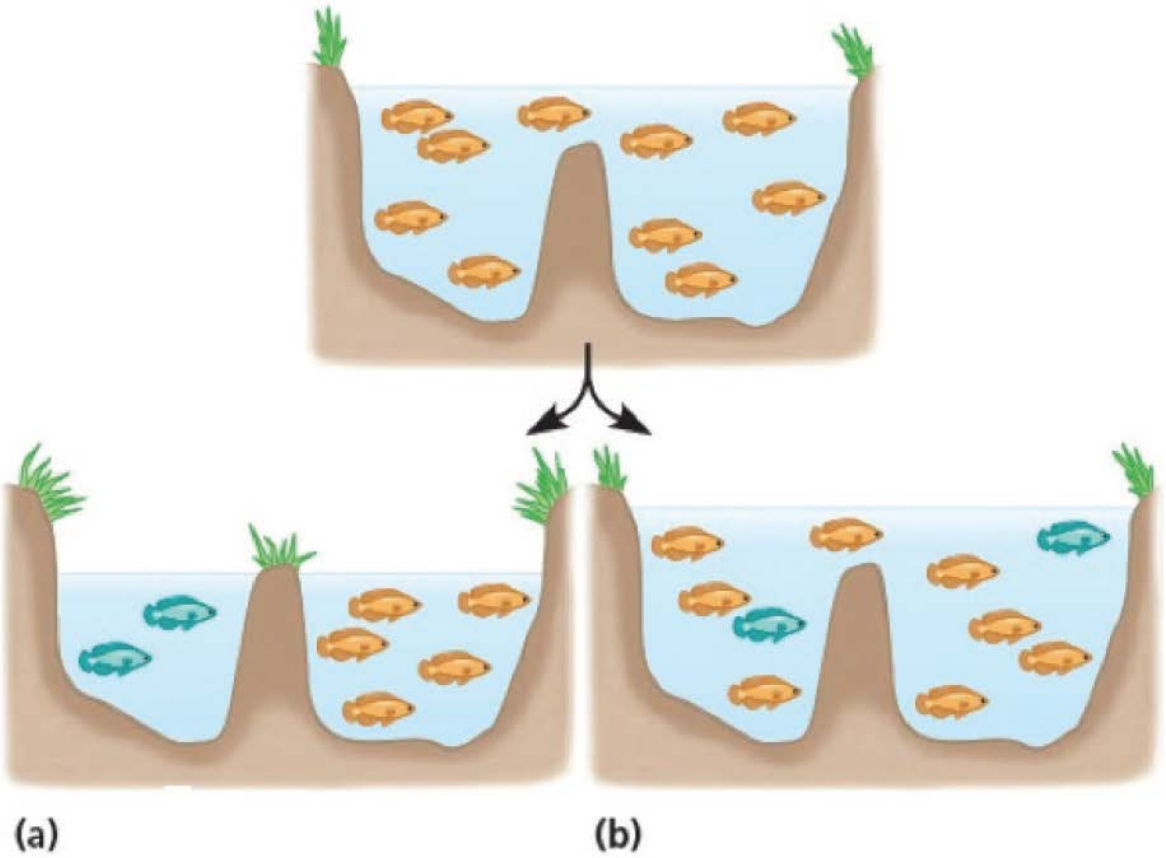
In a type of speciation called allopatric speciation organisms of one population are separated geographically and cannot reproduce. For example, when water level in a lake or sea decreases and it is divided into two, a population of fish is divided into two groups which cannot reproduce. If this division stays for a long period of time two groups of fish evolve separately, become different, and cannot reproduce any more. So, now we have two different species.

Another type of speciation is called sympatric speciation. Here new species are formed without geographic barriers by one of three factors:

1. Polyploidy - accidental increase in number of sets of chromosomes. Some polyploid organisms (mostly plants) can survive and form a new species, because they have different number of chromosomes now, and they cannot reproduce with original population;
2. Sexual selection - when females (or males) choose certain partners to mate. Having different preferences will make reproductive isolation, even if all organisms are in one place;
3. Habitat differentiation - differences in the source of food, or places to live, or time to mate in one population can also cause reproductive isolation, which leads to speciation.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ



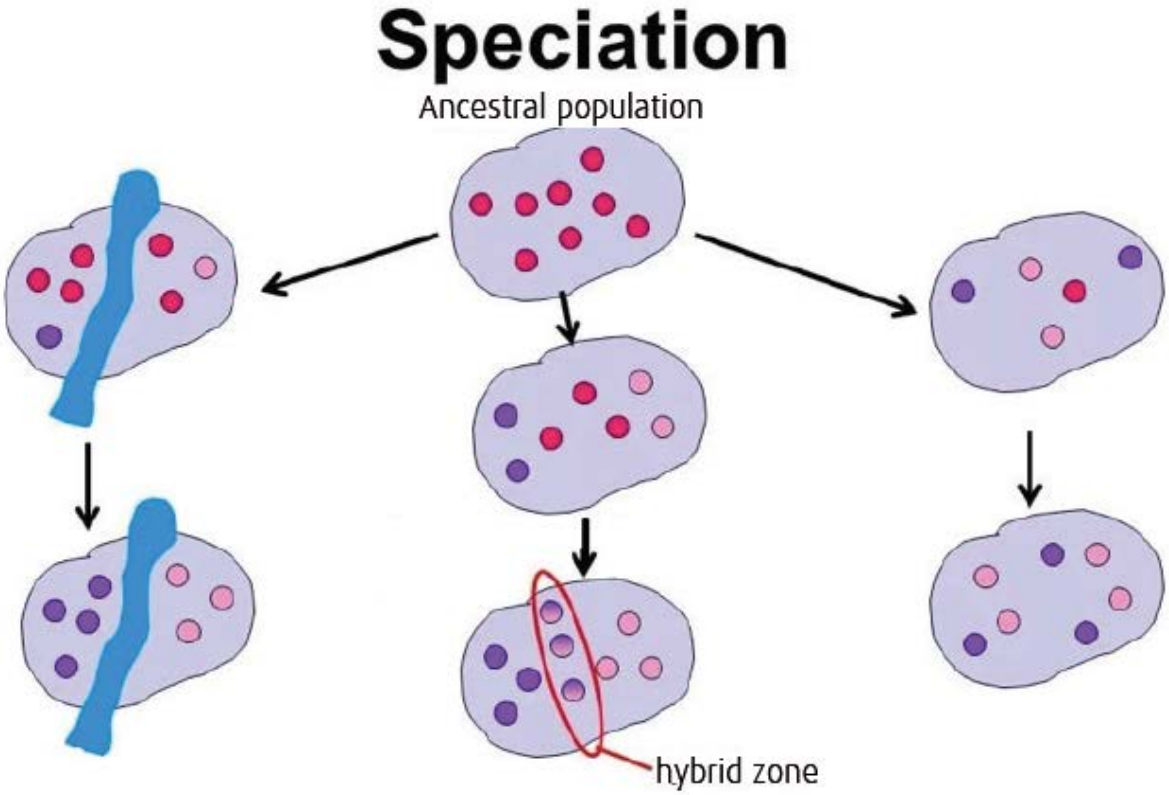
Two types of speciation: allopatric (a) and sympatric (b).

Activity

On the picture below you can see three types of speciation.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ



1. Find out allopatric speciation. Explain your answer.
2. Find out sympatric speciation. Explain your answer.
3. There is one more type of speciation called parapatric speciation. Using picture above explain parapatric speciation.

Facts

Intermediate species are evolutionary connection between two groups of organisms. For example, between fish and amphibia. According to evolution, they should have characteristics between previous group and emerging group. So, they would fail the competition with more adapted species. Mostly fossils of intermediate species are not found.

Research time

Найдите различные примеры видообразования. Определите тип видообразования и объясните почему.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

Literacy

1. Give three examples of changes in environment, which can cause allopatric speciation.
2. Although horse and donkey can mate and produce offspring called a mule, they are different species. Explain why?

Terminology

- ancestor - баба / предок;
- barrier - кедергі / барьер;
- certain - белгілі / определенный;
- criteria - белгі / критерий;
- fertile - өсімтал / плодовитый;
- isolation - оқшаулау / изоляция;
- mate - жұптасу / спариваться;
- mule - қашыр / мул;
- preference - артық көру / предпочтение;
- to separate - айырып алу / отделять.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

Problems

Test with one correct answer

1. Which one of these is an example of sympatric speciation?

A) the river began to flow through the valley, dividing it into two parts. A population of mice is also divided into two new populations

B) the emerging mountain divided the forest with a populations of pine trees into two

C) maggot flies divided according to their references in food

D) the water level in a lake lowers, dividing it into several new lakes. So, one population of fish is divided into many small populations

E) people built a railway in the steppe, and divided a population of earthworms into two parts

2. Which one of these is an example of natural selection?

A) water level raises divided one big land into many small islands

B) flood kills all insects in a valley

C) because of fire many trees die in a forest

D) people allow only some sheeps to reproduce

E) the fastest rabbit stay alive

3. Which one of these is not an example of Lamarckism?

A) if a man does gymnastics all his life, his son since birth will be slightly stronger than others

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

B) if you cut the tail for 20 generations of mice, mice of 21st generation will have a shorter tail

C) rabbits used their ears for many generations, that's why they have big ears

D) weak sparrows cannot reproduce, that's why there will be less weak sparrows

E) dogs have trained their sense of smell for thousands of years

4. In which period of time we are living right now?

A) proterozoic eon

B) archaen eon

C) cenozoic era

D) mesozoic era

E) paleozoic era

Test with several (max 3) correct answers

1. Which events happened in Proterozoic eon?

A) photosynthesis begins

B) primates emerged

C) the colonization of land

D) eukaryotic cells emerged

E) multicellular organisms emerge

F) the Origin of mammals

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

G) the first human appeared

H) the first animals appeared

2. What did Darwin do?

A) he first made a Classification system

B) he made a voyage around the Earth

C) he proposed the first theory of evolution

D) he published book “The Origin of Species”

E) he believed that all species do not change

F) he discovered the cell

G) he visited Kazakhstan

H) he was a teacher of Alfred Wallace

3. Which are the criteria of species?

A) organisms have the same chromosome number

B) organisms have the same colour

C) organisms can mate, but cannot produce offspring

D) organisms can mate, but cannot infertile offspring

E) organisms can mate, but cannot fertile offspring

F) organisms live in one place

G) organisms eat the same food

H) organisms have the same sizes

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

Matching questions (3 correct answers)

1. Match the following eras with events in these eras.

1. mesozoic era

2. cenozoic era

3. paleozoic era

A) colonization of land

B) the origin of Earth

C) mammals and birds spread

D) dinosaurs spread

E) the first prokaryotic cells emerged

F) the first eukaryotic cells emerged

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

ANSWERS

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

1. CELL BIOLOGY	1	2	3	4
Answers for the tests with one correct answer	C	B	E	C
Answers for the tests with several correct answers	C, D, H	C, G	A, C, G	
Matching questions	1-D, 2-A, 3-E	1-F, 2-C, 3-A		

2. BIOSPHERE	1	2	3	4
Answers for the tests with one correct answer	B	D	D	C
Answers for the tests with several correct answers	B, E, H	C, F	A, E, H	
Matching questions	1-B, 2-C, 3-E	1-D, 2-B, 3-A		

3. HUMAN IMPACT ON ENVIRONMENT	1	2	3	4
Answers for the tests with one correct answer	C	B	D	A
Answers for the tests with several correct answers	A, E, H	A, D, F	A, C, G	
Matching questions	1-D, 2-B, 3-E	1-D, 2-F, 3-C		

4. DIGESTION	1	2	3	4
Answers for the tests with one correct answer	C	B	C	C
Answers for the tests with several correct answers	D, E, G	F, G	C, G	
Matching questions	1-E, 2-C, 3-F	1-E, 2-A, 3-C		

5. TRANSPORT OF SUBSTANCES	1	2	3	4
Answers for the tests with one correct answer	A	C	D	B
Answers for the tests with several correct answers	A, C, G	A, E, H	B, D, G	
Matching questions	1-E, 2-A, 3-C	1-B, 2-E, 3-D		

6. RESPIRATION	1	2	3	4
Answers for the tests with one correct answer	A	C	D	C
Answers for the tests with several correct answers	B, E	B, E, F	A, D, E	
Matching questions	1-C, 2-F, 3-E	1-F, 2-D, 3-A		

7. EXCRETION	1	2	3	4
Answers for the tests with one correct answer	E	C	E	A
Answers for the tests with several correct answers	A, E, F	A, B	B, D, F	
Matching questions	1-D, 2-A, 3-B	1-D, 2-C, 3-B		

8. COORDINATION AND REGULATION	1	2	3	4
Answers for the tests with one correct answer	B	D	B	C
Answers for the tests with several correct answers	A, E, G	B, C, D	C, D, F	
Matching questions	1-D, 2-B, 3-F	1-A, 2-F, 3-C		

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

9. MOVEMENT	1	2	3	4
Answers for the tests with one correct answer	C	D	A	C
Answers for the tests with several correct answers	D, E, F	A, C, H	C, D, G	
Matching questions	1-C, 2-E, 3-B	1-F, 2-A, 3-E		

10. MOLECULAR BIOLOGY	1	2	3	4
Answers for the tests with one correct answer	A	D	C	B
Answers for the tests with several correct answers	B,D,F	A,D,G	A,D,F	
Matching questions	1-D, 2- A, 3-E	1-C, 2-B, 3-E		

11. CELL CYCLE	1	2	3	4
Answers for the tests with one correct answer	D	B	A	C
Answers for the tests with several correct answers	C,E,H	A,D,E	C,F,G	
Matching questions	1-D;2-F;3-E	1-B;2-C;3-D		

12. LAWS OF HEREDITY AND VARIATION	1	2	3	4
Answers for the tests with one correct answer	C	B	D	C
Answers for the tests with several correct answers	C,E,F	B,D,H	A,C,F	
Matching questions	1-C, 2-E, 3-A	1-A, 2-E, 3-D		

13. MICROBIOLOGY AND BIOTECHNOLOGY	1	2	3	4
Answers for the tests with one correct answer	E	B	A	C
Answers for the tests with several correct answers	D, H	B, C	A, D, G	
Matching questions	1-F, 2-D, 3-A	1-D, 2-A, 3-E		

14. HUMAN REPRODUCTION	1	2	3	4
Answers for the tests with one correct answer	C	A	D	B
Answers for the tests with several correct answers	A,G,H	D,G,H	A,C,F	
Matching questions	1-C, 2-D, 3-A	1-F, 2-E, 3-D		

15.GROWTH AND DEVELOPMENT	1	2	3	4
Answers for the tests with one correct answer	E	D	D	E
Answers for the tests with several correct answers	A, D, E	C, F	B, C, H	
Matching questions	1-D; 2-C; 3-A	1-E; 2-D; 3-B		

16. EVOLUTION	1	2	3	4
Answers for the tests with one correct answer	C	E	D	C
Answers for the tests with several correct answers	D, E, H	B, D	A, E	
Matching questions	1-D; 2-C; 3-A	1-A; 2-D; 3-C		

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

Gloassary

A

Absorption: uptake of substances by a tissue.

Action potential: the change in charges inside and outside of the cell.

Active transport: transport of large molecules with energy loss.

Adaptation: a process that fits organisms to their environment.

Aerobic respiration: the process of breakdown organic molecules using oxygen.

AIDS: acquired immunodeficiency syndrome.

Allele: versions of gene for one trait.

Ammonia: a strongly basic compound of nitrogen and hydrogen that is very soluble in water and formed in nature by-product of protein metabolism.

Amylase: enzyme, which breaks down carbohydrates.

Anaerobic respiration: the process of breakdown organic molecules without oxygen.

Antagonistic pair of muscles: a pair of muscles which have opposite functions.

Atmosphere: a layer of gases which surrounds the Earth.

Autosome: chromosome that carry genes for the body traits.

Auxin: plant hormone that regulates growth and other processes of plants.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

Abdomen: The part of the body of a vertebrate containing the digestive and reproductive organs; the belly.

Absorption: Take up or take in; The sponge absorbs water well

Acid: A chemical that reacts easily with other substances and turns litmus paper red; less than 7 on the pH scale

Acne: inflammatory disease of the sebaceous gland

Active immunity: is immunity in an organism resulting from its own production of antibody or lymphocytes

Agglutination: adhesion of separate parts

Albinism: Albinism is a range of disorders varying in severity. They are all caused by a reduction or absence of the pigment melanin, often causing white skin, light hair, and vision problems.

Alveoli: tiny air sacs found in a lung which is used for gas exchange

Amino Acid: A molecule that joins with other amino acids to form proteins

Analyze: Think about the different parts of a problem or situation to figure out how it is related to the whole.

Angiosperms: flowering plants

Arteriosclerosis: a disease of the arteries characterized by the deposition of fatty material on their inner walls.

Artery: Blood vessel that carries blood away from the heart

Arthropods: animals with segmented body, jointed limbs and exoskeleton

Asexual Reproduction: Reproduction that happens without sex; one organism creates one or more organisms

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

Atom: The smallest unit of a substance that has all of the properties of that substance

Atrium: upper chamber of heart that receives blood from the body and lungs

B

Binomial nomenclature: a system of nomenclature in which each species is given a unique name that consists of a generic and a specific term.

Biogeochemical cycle: the flow of chemical elements and compounds between living organisms and the environment.

Biotechnology: using of biological processes or organisms to produce different products.

Brain-Computer Interfaces: a system for exchanging information between the brain and an electronic device.

Backbone: the row of connected bones that go down the middle of the back and protect the spinal cord.

B cells: a kind of lymphocyte that provide humoral immunity

Bacteria: The kingdom of life which has no cell membrane or nucleus and is always unicellular

Beriberi: a disease causing inflammation of the nerves and heart failure, ascribed to a deficiency of vitamin B1.

Bile: a yellow or greenish liquid that is made by the liver and that helps the body to digest fats.

Biological classification: Organization of how living things are related to each other

Biology: The study of living things

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

Blastula: early stage of development, two-layered ball like structure

Body cell: All of the cells in an organism not involved in reproduction

Bond: An electrical force that links atoms together

Bronchi: major air passages of the lungs which diverge from the windpipe.

C

Canines: a pointed tooth between the incisors and premolars

Carbon: a widely distributed element that forms organic compounds in combination with hydrogen, oxygen, etc.

Carrying capacity: the number of organisms which a region can support without environmental degradation.

Cell cycle: it a period of cell life from one cell division to the next cell division.

Cellular respiration: the process of breakdown organic molecules producing energy.

Centrosome: an organelle, which plays role in cell division.

Cilia: a short structure, used for movement.

Classification: systematic arrangement in groups or categories according to established criteria.

Codominance: when the neither phenotype is dominant, the individual expresses both phenotypes.

Cystitis: inflammation of urinary bladder or urethra by infection of bacteria.

Canines: a pointed tooth between the incisors and premolars of a mammal.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

Calorie: Measure of energy, usually contained in food

Cancer: Disease where cells grow in an uncontrolled way

Capillaries are blood vessels that connect arteries and veins

Carbohydrate: An essential chemical in all cells that is broken down to form sugars; glucose, sucrose, lactose, galactose

Carbon Dioxide: Molecule made up of one carbon and two oxygens, produced by animals and other organisms; main contributor to manmade global warming

Cell: The basic structural and functional unit of all organisms

Cell differentiation: A process where cells change to perform different roles

Cell division: A process where one cell becomes more than one cell

Cell membrane: Surrounds the cell and keeps it together; also decides what material enters and leaves

Cell organelle: Parts of the cell that perform specific functions

Cell wall: The stiff outer layer of a cell that protects the cell and gives it shape

Cellular respiration: The process where organisms get energy from organic molecules

Cellulose: a substance that is the main part of the cell walls of plants and that is used in making various products (such as paper)

Characteristic: A distinguishing quality of something; generosity is one of his best characteristics

Chloroplast: Organelle in plants and some other organisms which is responsible for photosynthesis

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217

Все учебники Казахстана на OKULYK.KZ

Chest cavity: internal space of chest where heart and lungs are located.

Chitin: a kind of polysaccharide which is found in the structure of fungi cell walls and insect skeleton.

Closed circulatory system: blood is contained inside blood vessels, circulating unidirectionally.

Compound: A chemical combination of two or more atoms (of different elements)

Consumer: Living thing that eats other living things

Cotyledon: part of the seed that will grow into the leafs.

Covalent Bond: A type of chemical bond where electrons are shared between the atoms

Crop: organ used for storage of food in birds.

D

Deforestation: the cutting down of trees without replacing them.

Digestion: breaking down big molecules into small molecules.

DNA: a molecule that store and transmit genetic information.

Double helix: structure formed by two strands of DNA.

Dynamic work of muscles: muscle contract and relax to move different parts of your body.

Decomposer: A type of living thing that survives by consuming dead organic matter

Denaturation: change of the structure of protein by heat or an acid.

*Книга предоставлена исключительно в образовательных целях
согласно Приказа Министра образования и науки Республики Казахстан от 17 мая 2019 года № 217